

A Global Poincaré Inequality on the Finite-Type Model Boundary

Independent research note

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Abstract

Hytönen asks whether the finite-type model boundary $\partial\Omega_k$ from Chang–Li–Ottazzi–Wu satisfies the complete metric $(1, 4)$ -Poincaré assumption. The answer is yes. Its horizontal fields are homogeneous Hörmander fields, so a global theorem of Biagi–Bonfiglioli–Bramanti gives the stronger $(1, 1)$ inequality with one constant for all control balls. We verify the homogeneity and rank hypotheses, pass from smooth horizontal gradients to the lower pointwise slope of arbitrary Lipschitz functions, and prove completeness. Hence the source theorem applies without the previously imposed C^2 assumption.

Packet status

Run	fa_banach_001, lane 13.
Source	arXiv:2411.02613, Example 1.24, PDF page 8 [1].
Classification	Candidate full positive answer, likely valid subject to human mathematical review.
Novelty	Low-to-moderate. The global theorem is known, but no explicit application answering this question was located.

1.24. *Example (plus Open Problem).* The setting of [11] involves a space of homogeneous type $\partial\Omega_k$ of lower dimension $d = 4$ and upper dimension $D = 2k + 2$, where $k \geq 2$; see the penultimate paragraph of [11] Introduction]. For these values, [11] Theorem 1.3] gives the characterisation (1.15), which agrees with the numerology of our setting. However, the following question is open to our knowledge:

Does $\partial\Omega_k$ of [11] satisfy the Poincaré assumption (iii)?

If the answer is positive, this would have the advantage of not only recovering [11] Theorem 1.3] but also lifting the *a priori* assumption $b \in C^2(\partial\Omega_k)$ in part (2) of the said result, since no such condition appears in our Theorem 1.1

1 Geometry and precise answer

Identify $\partial\Omega_k$ with $\mathbb{R}^3$ in coordinates (x, y, t) , and put $r_0^2 = x^2 + y^2$. The CR horizontal fields are

$$X_1 = \partial_x + yr_0^{2k-2}\partial_t, \quad X_2 = \partial_y - xr_0^{2k-2}\partial_t, \quad k \geq 2. \quad (1)$$

Let d_{cc} be their Carnot–Carathéodory distance, let $\mu = dx dy dt$, and write $B(p, r) = B_{d_{cc}}(p, r)$. For a d_{cc} -Lipschitz function define, exactly as in the source,

$$\text{lip } f(p) = \liminf_{s \downarrow 0} \sup_{d_{cc}(p, q) \leq s} \frac{|f(p) - f(q)|}{s}.$$

Theorem 1. *For every integer $k \geq 2$, $(\partial\Omega_k, d_{cc}, \mu)$ is a complete doubling metric measure space and there is a constant C_k such that every d_{cc} -Lipschitz f satisfies*

$$\int_{B(p,r)} |f - f_{B(p,r)}| d\mu \leq C_k r \left(\int_{B(p,2r)} (\text{lip } f)^4 d\mu \right)^{1/4} \quad (2)$$

for all $p \in \partial\Omega_k$ and $r > 0$. In fact the stronger $(1,1)$ inequality

$$\int_{B(p,r)} |f - f_{B(p,r)}| d\mu \leq C'_k r \int_{B(p,2r)} \text{lip } f d\mu \quad (3)$$

holds.

This is precisely the Poincaré assumption in the question, with lower dimension $d = 4$ and dilation factor $\lambda = 2$.

Remark 1 (Two distances in the cited paper). Chang–Li–Ottazzi–Wu also denote by d the quantity $h(p,q)^2 \rho(p,q)^{2-2k}$. Their later identity is

$$d(p,q) \asymp \mu(B_{d_{cc}}(p, d_{cc}(p,q))),$$

so this d is a volume gauge. The length metric having lower dimension 4 and upper dimension $2k + 2$ is d_{cc} . Indeed their ball estimate is

$$\mu(B((z,t), r)) \asymp r^4(|z| + r)^{2k-2}. \quad (4)$$

Consequently, for $0 < r \leq R$,

$$(R/r)^4 \lesssim \frac{\mu(B(p,R))}{\mu(B(p,r))} \lesssim (R/r)^{2k+2}.$$

This also explains exactly the dimensions quoted in the open problem [2].

2 The global inequality

Consider the anisotropic dilations

$$\delta_s(x, y, t) = (sx, sy, s^{2k}t).$$

Both fields in (1) are homogeneous of degree one:

$$X_j(g \circ \delta_s) = s(X_j g) \circ \delta_s, \quad j = 1, 2. \quad (5)$$

They are linearly independent as vector fields, since their ∂_x and ∂_y components are distinct. They also satisfy Hörmander's rank condition at the origin. Directly,

$$[X_1, X_2] = -2k(x^2 + y^2)^{k-1} \partial_t. \quad (6)$$

Since the coefficient in (6) is independent of t , bracketing another $2k - 2$ times with X_1 simply differentiates that coefficient in x . Hence

$$(\text{ad } X_1)^{2k-2}[X_1, X_2](0) = -2k(2k-2)! \partial_t. \quad (7)$$

Together with $X_1(0) = \partial_x$ and $X_2(0) = \partial_y$, this spans $T_0\mathbb{R}^3$.

Theorem 7.6 of Biagi–Bonfiglioli–Bramanti states that every family of degree-one homogeneous fields satisfying this rank condition obeys the global estimate

$$\oint_{B(p,r)} |u - u_{B(p,r)}| d\mu \leq Cr \oint_{B(p,2r)} |Xu| d\mu, \quad |Xu| = (|X_1u|^2 + |X_2u|^2)^{1/2}, \quad (8)$$

for every $p \in \mathbb{R}^3$, every $r > 0$, and every C^1 function near $B(p, 2r)$ [3]. The point of that theorem is global uniformity: its constant is independent of both p and r . Equations (5)–(7) verify all of its hypotheses.

As a last application, in order to have the following Harnack Theorem [7.7](#) we need to check the validity of the next axiom in our framework:

(A.9) The global doubling inequality [2.3](#) holds, together with the following result:

Theorem 7.6 (Global Poincaré inequality). *Let $X = \{X_1, \dots, X_m\}$ satisfy axioms (H.1) and (H.2). There exists a constant $C_P > 0$ such that, for every $x \in \mathbb{R}^n$, $r > 0$ and every u which is C^1 in a neighborhood of $B_X(x, 2r)$, one has*

$$(7.7) \quad \oint_{B_X(x,r)} |u(y) - u_{B_X(x,r)}| dy \leq C_P r \oint_{B_X(x,2r)} |Xu(y)| dy.$$

As usual we have set $|Xu| := \sqrt{\sum_{j=1}^m |X_j u|^2}$, and (if B is any d_X -ball)

$$u_B := \oint_B u := \frac{1}{|B|} \int_B u(y) dy.$$

3 Lipschitz slope and completeness

It remains to match (8), stated first for smooth functions, with the lower metric slope in the question.

Lemma 1. *If f is d_{cc} -Lipschitz, then its horizontal weak derivatives exist and*

$$|X_j f| \leq \text{lip } f \quad \text{a.e.}, \quad j = 1, 2.$$

Consequently $|Xf| \leq \sqrt{2} \text{lip } f$ almost everywhere.

Proof. Let Φ_j^s be the flow of X_j . It is a horizontal curve of speed one, so

$$d_{cc}(p, \Phi_j^s(p)) \leq |s|.$$

Thus $s \mapsto f(\Phi_j^s(p))$ is Lipschitz. The one-dimensional differentiation theorem in flow boxes, followed by Fubini, gives $X_j f$ for almost every p . At every such point,

$$\begin{aligned} |X_j f(p)| &= \lim_{s \downarrow 0} \frac{|f(\Phi_j^s(p)) - f(p)|}{s} \\ &\leq \liminf_{s \downarrow 0} \sup_{d_{cc}(p,q) \leq s} \frac{|f(q) - f(p)|}{s} = \text{lip } f(p). \end{aligned}$$

The final assertion follows by taking the Euclidean norm of the two horizontal derivatives. $\square$

Control-metric Lipschitz functions belong locally to $W_X^{1,\infty}$. Standard smooth density for Hörmander Sobolev spaces allows (8) to pass to them in $W_X^{1,1}$ on each doubled ball [4]. Lemma 1 therefore gives (3). Finally, Hölder’s inequality on the probability space $(B(p, 2r), \mu/\mu(B(p, 2r)))$ gives (2).

For completeness, let γ be a horizontal curve of length L starting at (z_0, t_0) , parametrized with control speed at most one. From (1),

$$|z(s) - z_0| \leq L, \quad |t(s) - t_0| \leq L(|z_0| + L)^{2k-1}.$$

Thus every bounded d_{cc} -ball is Euclidean bounded. The ball-box theorem for bracket-generating smooth fields says that the d_{cc} and Euclidean topologies agree; since d_{cc} is continuous, closed bounded control balls are Euclidean compact. Hence $(\mathbb{R}^3, d_{cc})$ is proper, and therefore complete. This finishes the proof of Theorem 1.

4 Consequence and verification boundary

The answer is stronger than requested: no exponent-specific argument is needed, since (1, 1) implies (1, p) for every $p \geq 1$. Combined with the source theorem, the positive answer removes the a priori $b \in C^2(\partial\Omega_k)$ condition in the cited low-index Schatten result.

A bounded literature search on August 17, 2026 covered the exact question, the three relevant arXiv identifiers, the model fields, and combinations of “global Poincaré”, “homogeneous Hörmander”, and $\partial\Omega_k$. It found the global theorem used above but no later paper explicitly answering the source question. The computation of (6)–(7) and the homogeneity check were also verified symbolically for $k = 2, \dots, 8$. The main human-review point is the notation disambiguation in Remark 1; the dimension statement in the question and CLOW’s own volume identity both select d_{cc} .

References

- [1] T. Hytönen, *Schatten properties of commutators on metric spaces*, arXiv:2411.02613.
- [2] D.-C. Chang, J. Li, A. Ottazzi, and Q. Wu, *Optimal lifting of Levi-degenerate hypersurfaces and applications to the Cauchy–Szegő projection*, arXiv:2309.00525; Adv. Math. 496 (2026), 111000.
- [3] S. Biagi, A. Bonfiglioli, and M. Bramanti, *Global estimates for the fundamental solution of homogeneous Hörmander operators*, Ann. Mat. Pura Appl. 201 (2022), 1875–1934, Theorem 7.6; arXiv:1906.07836.
- [4] S. Biagi and M. Bramanti, *Global Sobolev regularity for nonvariational operators built with homogeneous Hörmander vector fields*, J. Differential Equations 423 (2025), 708–764, Theorem 1.3(a); arXiv:2312.15367.